Kvadratické rovnice a nerovnice - B

1. Řešte v R: $a)\ (x+\sqrt{7})(2x-5)=0$ $\quad b)\ x^2-8=0$ $\quad c)\ x^2+x\sqrt{11}=0$

$K=\{-\sqrt{7}\cdot\frac{5}{2}\}$

$(x-\sqrt{8})(x+\sqrt{8})=0$

$\sqrt{8}=2\sqrt{2}$

$K=\{\pm\sqrt{8}\}$

$x(x+\sqrt{11})=0$

$K=\{0;-\sqrt{11}\}$

2. Řešte užitím Vietových vzorců:

$a) x^2+10x+16=0$

$x_1+x_2=-b$

$x_1\cdot x_2=c$

$(x+8)(x+2)=0$

$-8 \quad -2$

$x_1+x_2=-10$

$x_1\cdot x_2=16 \rightarrow$

$4,4$ ✗ $\quad 1,16$

$-4,-4$ ✗ $\quad -1,-16$

$8,2$ ✗

$-8,-2$ ✓

$x_1=-8$

$x_2=-2$

$b)\ x^2-14x-15=0$

$-1,15$ ✓

$1,-15$

$5,-3$

$-5,3$

$x_1=-1$

$x_2=15$

$(x+1)(x-15)=0$

3. Určete číslo c, je-li $x_1=-9,\quad x^2+12x+c=0$

$x_1+x_2=-12$

$x_1\cdot x_2=c$

$-9+x_2=-12$

$x_2=-3$

$x_1\cdot x_2=c$

$-9\cdot(-3)=c$

$c=27$

4. Řešte v R:

$a)\ x^2+6x+13<0$

$D=36-4\cdot 13=36-52=-16$ nebere rozlišit

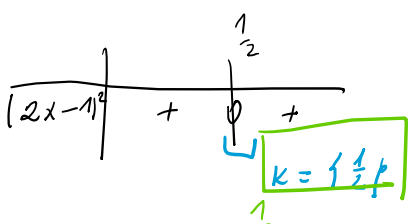
$x=0: \quad 0^2+6\cdot 0+13<0$

$13<0$ není pravda $\rightarrow$ $K=\emptyset$

$b)\ 4x^2-4x+1\le 0$

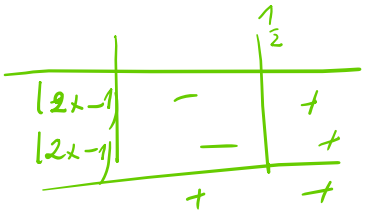
$(2x-1)^2\le 0 \quad \Leftrightarrow D=0$

$x=\frac{1}{2}$



$K=\{\frac{1}{2}\}$

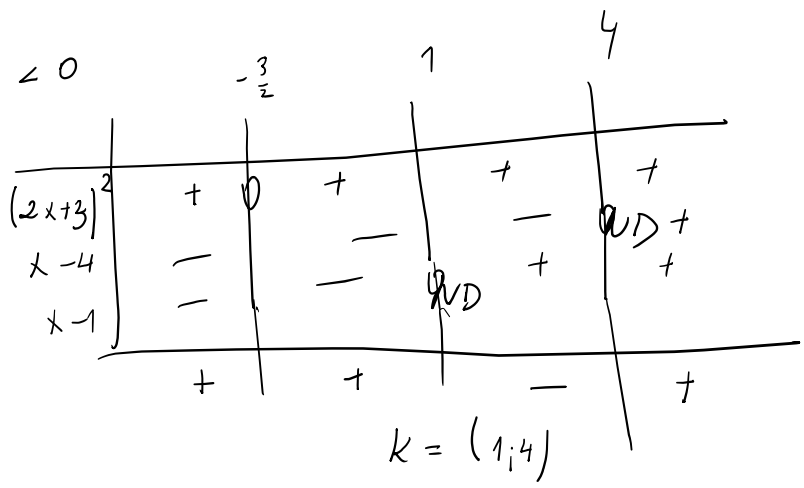
nebo



5. $\frac{4x^2+12x+9}{x^2-5x+4} < 0$

$-\frac{3}{2}; 1; 4$

$\frac{(2x+3)^2}{(x-4)(x-1)} < 0$



$K = (1;4)$

6. Sestavte kvadratickou rovnici, která má dvojnásobný kořen $1 + \sqrt{2}$.

$\downarrow$

$x_1 = x_2 \quad (D = 0)$

$(x - x_1)(x - x_1) = 0$

$[x - (1+\sqrt{2})]^2 = 0$

$x_1 + x_1 = -b$

$1+\sqrt{2} + 1+\sqrt{2} = -b$

$2 + 2\sqrt{2} = -b$

$b = -2 - 2\sqrt{2}$

$x_1 \cdot x_2 = c$

$(1+\sqrt{2})^2 = c$

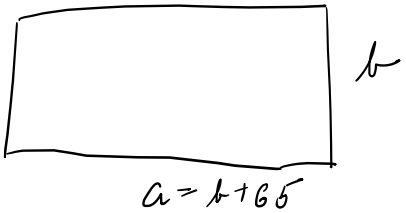
$1 + 2\sqrt{2} + 2 = c$

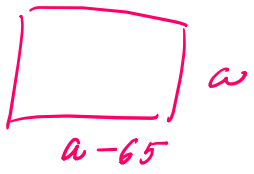
$3 + 2\sqrt{2} = c$

$a[x^2 + x(-2-2\sqrt{2}) + 3 + 2\sqrt{2}] = 0 \qquad a \neq 0$

7. Určete rozměry obdélníka, jehož obsah je 6 084cm^2 a jehož délka je o 65 cm větší než šířka.

$S = ab$





$b \cdot (b + 65) = 6084$

$b^2 + 65b - 6084 = 0$

$b_1 = 52 \Rightarrow a = b + 65 = 117$

$b_2 = -117$ ✗

$a_1 = 117 \Rightarrow b = 52$

$a_2 = -52$ ✗

Kvadratické rovnice a nerovnice – shrnutí B

1. Řešte v R:

$a)\left(x-\sqrt{7}\right)\left(3x+\sqrt{3}\right)=0$ $b)\ \sqrt{6}-x^2=0$ $c)x^2+\frac{x}{4}=0$

2. Řešte: $x^2-9x+8=0$

3. Sestavte kvadratickou rovnici o kořenech $x_1=3\sqrt{3}; x_2=-2\sqrt{3}$

4. $21-29x\leq 2(3-2x)^2$

5. $-3x^2 < -2$